KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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Course Code : EE 4121

VLSI

BiCMOS Logic

• BiCMOS Logic = Bipolar technology + CMOS technology
•BiCMOS => High speed or faster (BJT) & low power dissipation (CMOS)
•Speed and power product will be good.
•In BiCMOS=> MOS switches to perform the logic function and BJT to drive the output loads.

• The output logic levels will be good and will be close to the rail voltages since V_{CEsar} is quite small and V_{BE} is approximately + 0.7 volts.
• The inverter has a high input impedance.
• The inverter has a low output impedance.
• The inverter has a high current drive capability but occupies a relatively small area.
• The inverter has high noise margins.

BiCMOS Logic

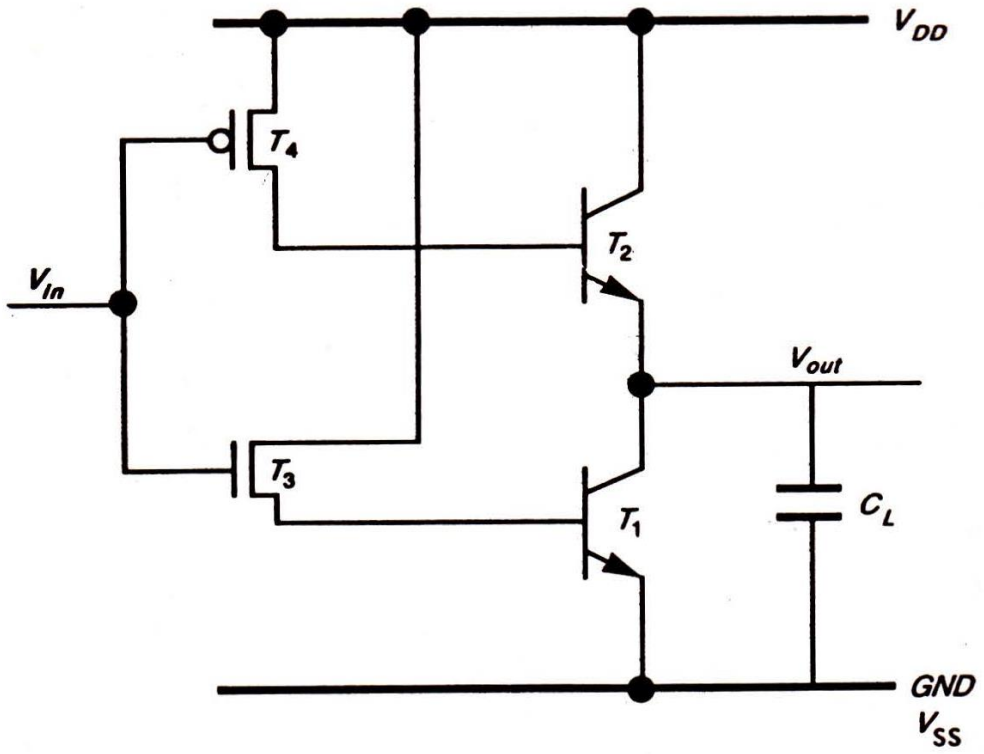


FIGURE 2.17 A simple BiCMOS inverter.

- BiCMOS Logic=> Just same as Inverter

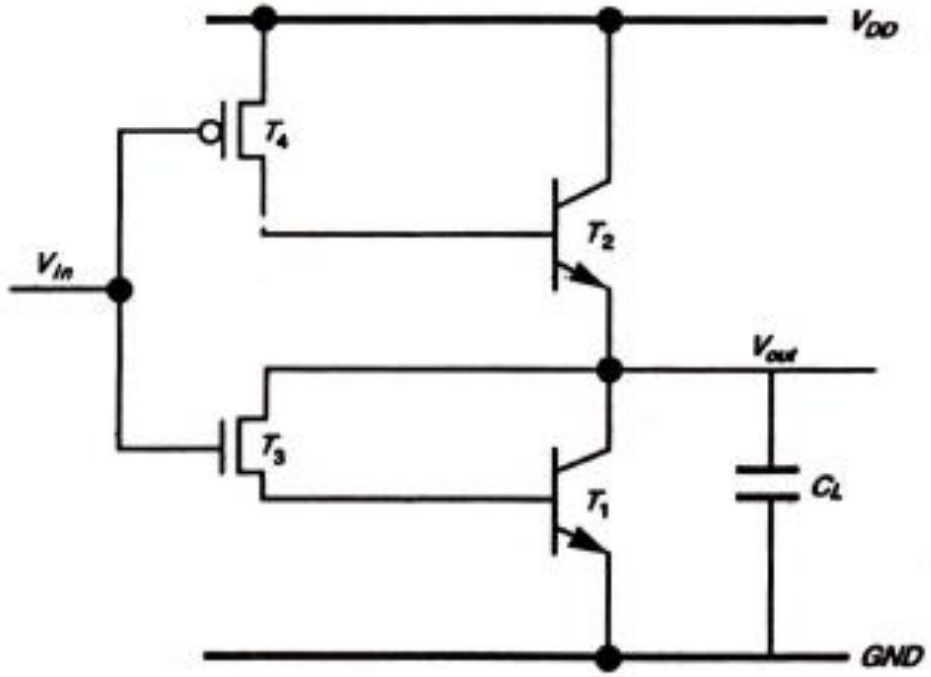
- When Vin = 1, then T4 and T2 off and T3 and T1 on and close then Vout will be 0

- When Vin = 0, then T4 and T2 on and T3 and T1 off, So output will be 1.

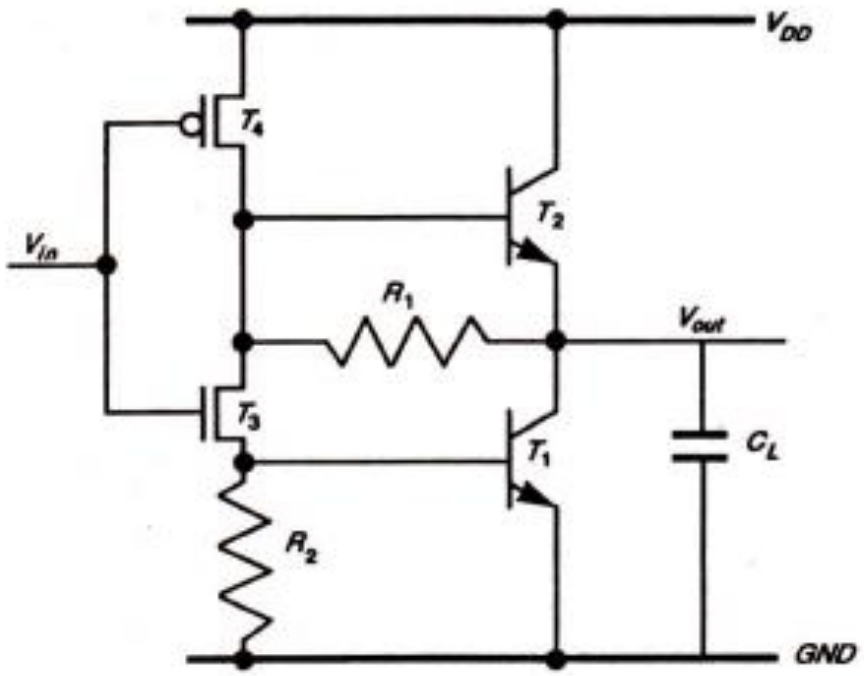
- Problems: When Vin = 1, then T3 and T1 on, so static current flow which is not good to implement this.

- There is also a problem in that there is not discharge path for current from the base of either BJT when it is being turned off. This will slow down the action of this circuit.

Modified BiCMOS Logic



E 2.18 An alternative BiCMOS inverter with no static current flow.

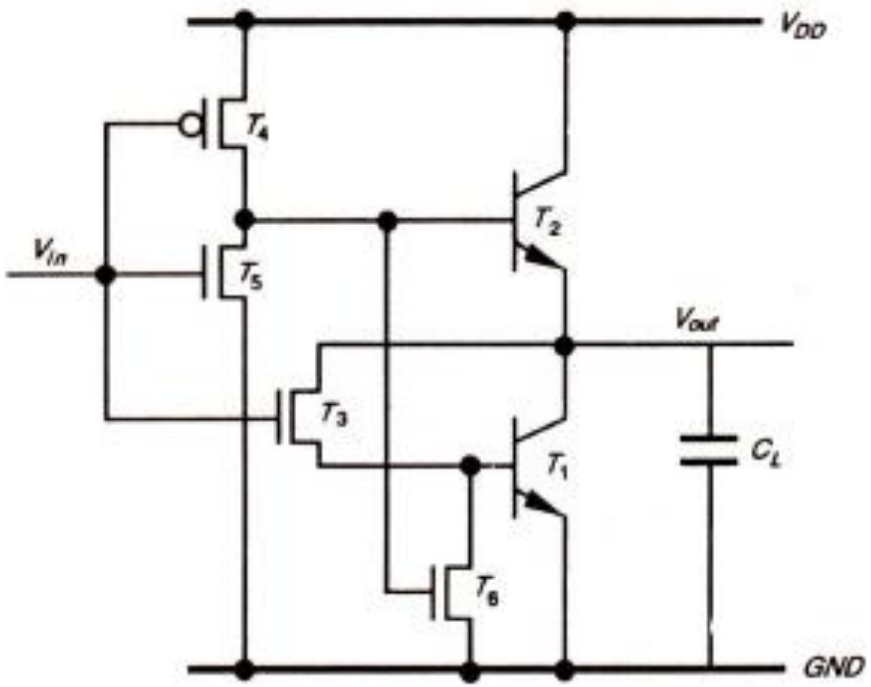


tE 2.19 An improved BiCMOS inverter with better output logic levels.

An improved version of this circuit is given in Figure 2.18, in which the DC path through T3 and T1 is eliminated, but the output voltage swing is now reduced, since the output cannot fall below the base to emitter voltage V8E of T1.

An improved inverter arrangement, using resistors, is shown in Figure 2.19. In this circuit resistors provide the improved swing of output voltage when each bipolar transistor is off, and also provide discharge paths for base current during turn-off.

Improved BiCMOS Logic



2.20 An improved BiCMOS inverter using MOS transistors for base current dischange.

The provision of on chip resistors of suitable value is not always convenient and may be space-consuming, so that other arrangements-such as in Figure 2.20-are used. In this circuit, the transistors T_5 and T_6 are arranged to turn on when T_2 and T_1 respectively are being turned off.

Latch-up in CMOS

A latch-up is a type of short circuit which can occur in an IC.
More specially, it is the low impedance path between Vdd and Vss of a MOS circuit due to interaction of parasitic pnp and npn bipolar transistors.

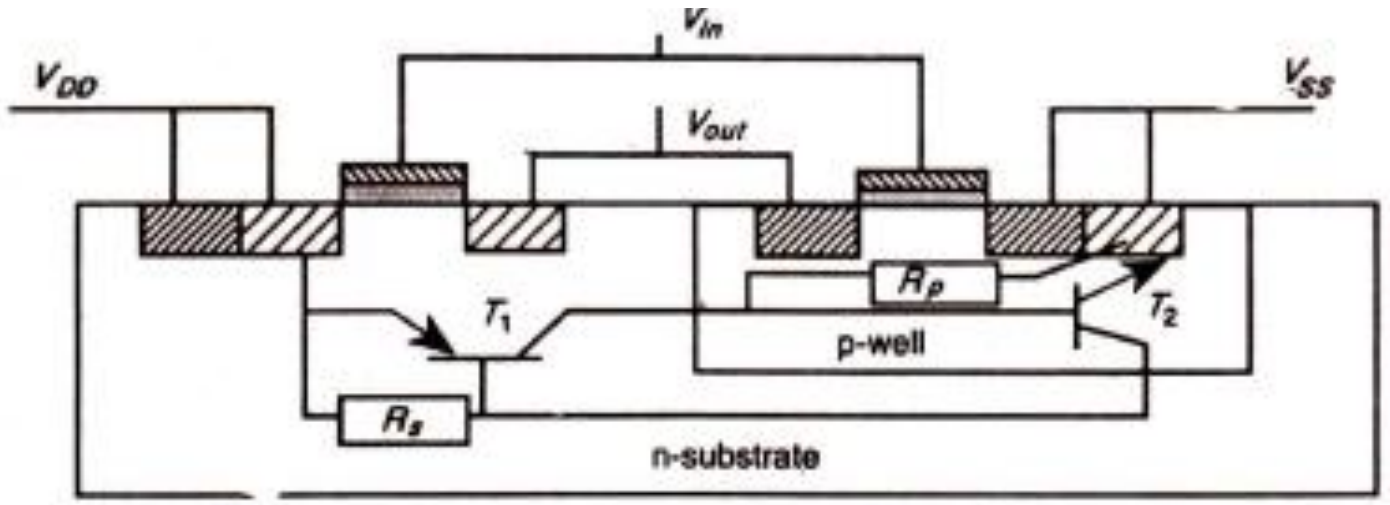


FIGURE 2.21 Latch-up effect in p-well structure.

Latch-up may be induced by glitches on the supply rails or by incident radiation. The mechanism involved may be understood by referring to Figure 2.21, which shows the key parasitic components associated with a p-well structure in which an inverter circuit (for example) has been formed.

Latch-up in CMOS

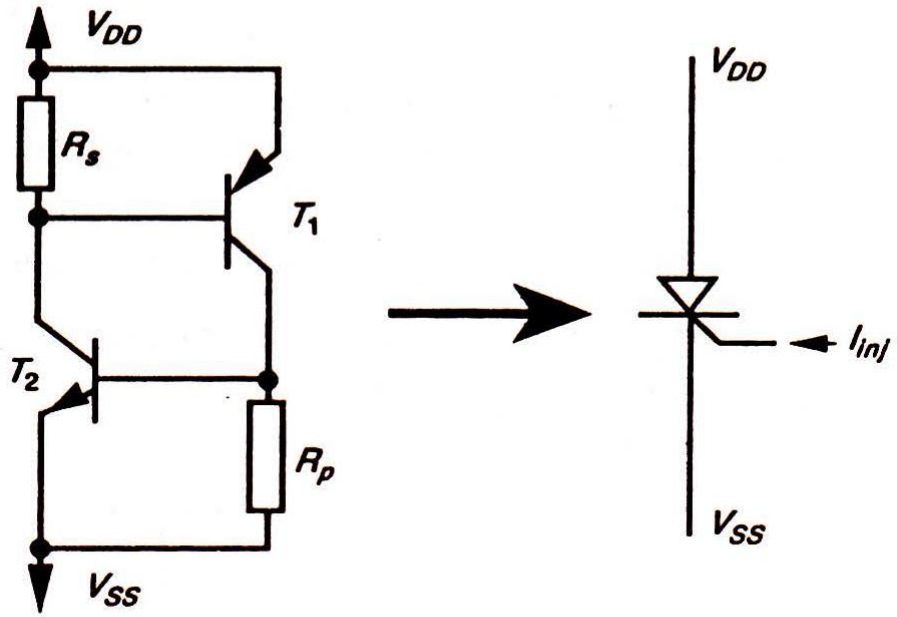


FIGURE 2.22 Latch-up circuit model.

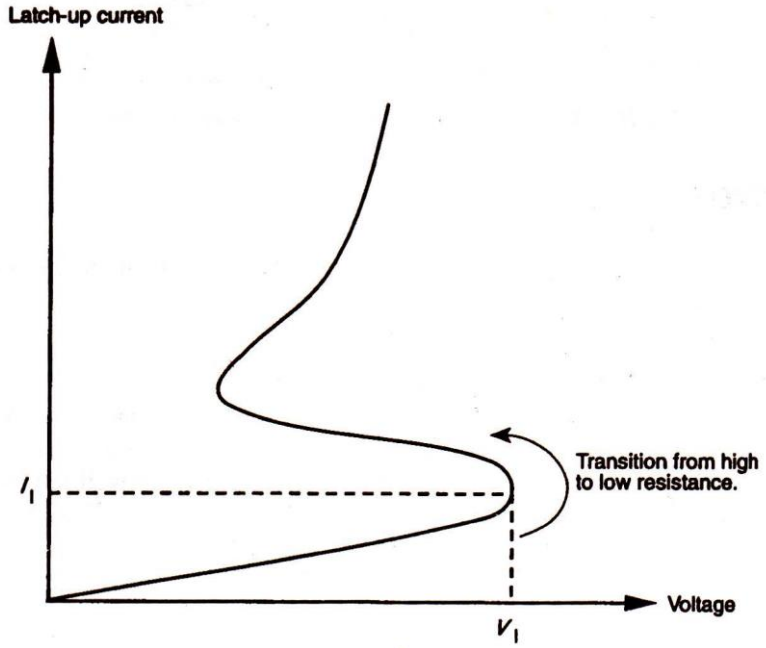


FIGURE 2.23 Latch-up current versus voltage.

To minimize latch-up:

- **Reduce Rs by increasing doping in substrate**
- **Reduce Rp by controlling fabrication**
- **Use guard ring**

With no injected current, the parasitic transistors will exhibit high resistance, but sufficient substrate current flow will cause switching to the low-resistance state as already explained. The switching characteristic of the arrangement is outlined in Figure 2.23.

A **guard ring works** by reducing the potential difference (voltage) to nearly zero, between the active (signal) conductor and the active conductor's surroundings. The **guard ring** is made to be "the surroundings". When the voltage is zero, no current can flow.

Pseudo nMOS logic

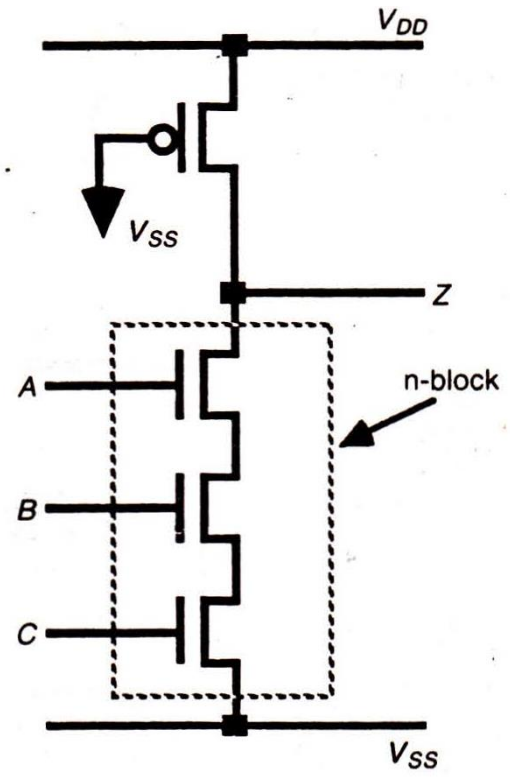


FIGURE 6.9 Pseudo-nMOS *Nand* gate.

Clearly, if we replace the depletion mode pull-up transistor of the standard nMOS circuits with a p-transistor with gate connected to *Vss*, we have a structure similar to the nMOS equivalent. This approach to logic design is illustrated by the three-input *Nand* gate in Figure 6.9. The circuit arrangements look and behave much like nMOS circuits and appropriate ratio rules must be applied.

Pseudo nMOS Inverter

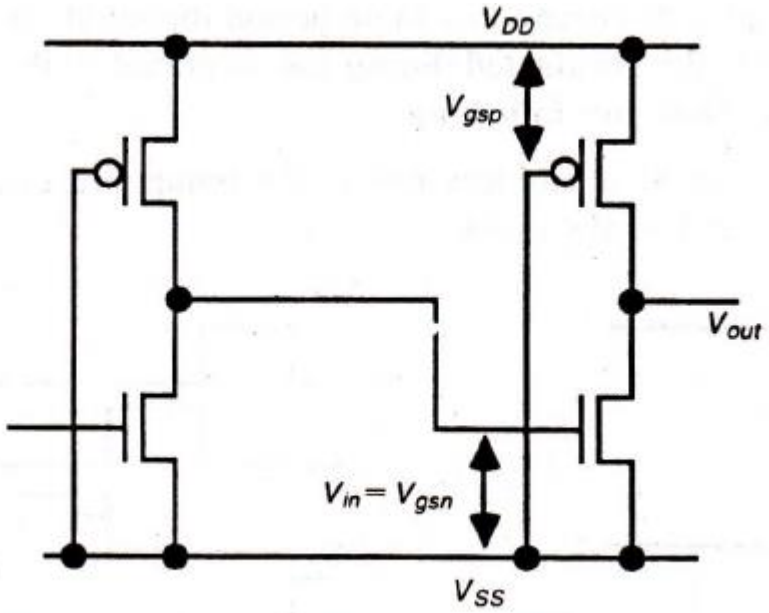


FIGURE 6.10 Pseudo-nMOS inverter when driven from a similar inverter.

In order to determine the required ratio, we consider the arrangement of Figure 6.10 in which a pseudo-nMOS inverter is being driven by another similar inverter, and we consider the conditions necessary to produce an output voltage of V_{inv} for an identical input voltage. As for the nMOS analysis, we consider the conditions for which $V_{inv} = V_{DD}/2$.

At this point the n-device is in saturation (i.e. $0 < V_{gsn} - V_{tn} < V_{dsn}$) and the p-device is operating in the resistive region (i.e. $0 < V_{dsp} < V_{gsp} - V_{tp}$). Equating currents of the n-transistor and the p-transistor, and by suitable rearrangement of the resultant expression, we obtain

Pseudo nMOS Inverter

$$V_{inv} = V_{tn} + \frac{(2\mu_p/\mu_n)^{1/2}[(-V_{DD} - V_{tn})V_{dsp} - V_{dsp}^2]^{1/2}}{(Z_{p.u.}/Z_{p.d.})^{1/2}}$$

where

$$Z_{p.u.} = L_p/W_p$$

and

$$Z_{p.d.} = L_n/W_n$$

With

$$V_{inv} = 0.5V_{DD}$$

$$V_{tn} = |V_{tp}| = 0.2V_{DD}$$

$$V_{DD} = 5\ \text{V}$$

$$\mu_n = 2.5\ \mu_p$$

we obtain

$$\frac{Z_{p.u.}}{Z_{p.d.}} = \frac{3}{1}$$

A transfer characteristic, V_{out} vs V_{in}, can be drawn and, as for the nMOS case, the characteristic will shift with changes of $Z_{p.u.}/Z_{p.d.}$ ratio.

Two points require comment:

1. Since the channel sheet resistance of the p-pull-up is about 2.5 times that of the n-pull-down, and allowing for the ratio of 3:1, the pseudo-nMOS inverter presents a resistance between V_{DD} and V_{SS} which is, say, 85 kΩ compared with 50 kΩ for a comparable 4:1 nMOS device. Thus, power dissipation is reduced to about 60% of that associated with the comparable nMOS device.
2. Owing to the higher pull-up resistance, the inverter pair delay is larger by a factor of 8.5:5 than the 4:1 minimum size nMOS inverter.